

第三章 遗传的染色体学说与连锁分析

The Chromosome Theory of Inheritance and Linkage Analysis

1.（改编自 D5-7, 2nd ed）人类常染色体等位基因 N 能导致指甲和髌骨的异常，称为指甲—髌骨综合征（nail patella syndrome）。一个患此病的 A 型血的人和—个正常的、具 O 型血的人结婚，他们的孩子中有 A 型血的该病患者。假定一些没有亲缘关系、但都具有这种表型的孩子（F）长大并且通婚，在他们的孩子们（F，第二代子女）中，表型及其比例如下：

- 甲—髌综合征，A 型 66%
- 正常，O 型 16%
- 正常，A 型 9%
- 甲—髌综合征，O 型 9%

分析上述数据，解释上述表型频率产生的原因。

答：

2.（改编自 D5-8, 2nd ed）有一个果蝇品系，对常染色体隐性等位基因 a、b、c 是纯合的，并且这 3 个基因以 a、b、c 的顺序连锁。将这种雌性果蝇和野生型雄果蝇杂交得到 F，F 相互杂交得到 F2 如下：

- （1）a 与 b，b 与 c 之间的重组率是多少？
- （2）并发系数是多少？

+	+	+	1364
a	b	c	365
a	b	+	87
+	+	c	84
a	+	+	47
+	b	c	44
a	+	c	5
+	b	+	4

答：

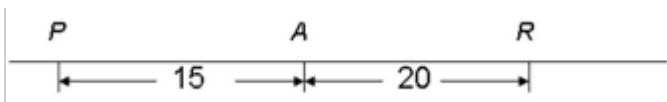
3. (改编自 D5-9, 2nd ed) 玉米的第三染色体有 3 个基因座, 分别是等位基因 b^+ 和 b 、 v^+ 和 v 、 y^+ 和 y 。加号“+”表示野生型表型, 与之相对应的隐性表型简写为 b (深绿色叶片)、 v (淡绿色叶片)、 y (无叶舌)。将对 3 个基因杂合的 F_1 植株与对 3 个基因隐性纯合的植株进行测交, 后代的表型如右侧所示:

计算各基因之间的图距, 作染色体图, 求并发系数。

$+$	v	y	305
b	$+$	$+$	275
b	$+$	y	128
$+$	v	$+$	112
$+$	$+$	y	74
b	v	$+$	66
$+$	$+$	$+$	22
b	v	y	18

答:

3. (改编自 D5-11, 2nd ed) 假设 Spock 先生的父亲是来自火星的星际飞船“企业号”的船长, 而他的母亲是地球人。火星人有尖耳朵 (由等位基因 P 决定)、无肾上腺 (由等位基因 A 决定)、心脏靠右 (由等位基因 R 决定), 这 3 个基因对于地球上相应的基因呈显性。这 3 个基因位于常染色体上, 它们的连锁关系如下:

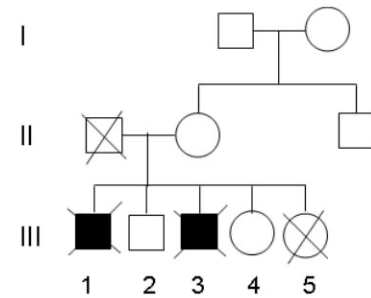


如果 Spock 先生与一位地球上的女子结婚, 不考虑干涉, 则他们的孩子中有多少:

- (1) 3 个性状都像火星人;
- (2) 3 个性状都像地球人;
- (3) 火星人的耳朵和心脏, 但是地球人的肾上腺;
- (4) 火星人的耳朵, 但是地球人的心脏和肾上腺。

答:

5. (改编自 D5-13, 2nd ed) 人的色盲基因和血友病基因都位于 X 染色体上, 它们之间的重组率大约是 10%。利用一个无害基因与致病基因的连锁关系, 可以做遗传诊断。这里给出的是某个系谱的一部分。黑色的符号表示该个体患有血友病, 叉号表示该个体患有色盲症。个体 III-4 和 III-5 的儿子患血友病的概率有多大?



答:

6. (Adapted from H3e, 5-6) We have applied the chi-square method to test linkage between genes *A* and *B* by asking whether the observed numbers of parental and recombinant classes differed significantly from the expectation of independent assortment that parentals = recombinants, based on the two sets of experimental data obtained from test cross below (*AB* and *ab* are parental traits, while *Ab* and *aB* are recombinants.) Another possible way to analyze the results from these same experiments is to ask whether the observed frequencies of the four genotypic classes (*AB*, *ab*, *Ab* and *aB*) can be explained by a null hypothesis predicting that they should appear in a 1 : 1 : 1 : 1 ratio. In order to consider the relative advantages and disadvantages of analyzing the data in these two different ways answer the following:

- What is the null hypothesis in each case?
- Which is a more sensitive test of linkage? (Analyze the data by second method, *i. e.*, according to the prediction of a 1 : 1 : 1 : 1 ratio.)

Progeny	Experiment 1		Experiment 2	
<i>A B</i>	17		34	
<i>a b</i>	14		28	
<i>A b</i>	8		16	
<i>a B</i>	11		22	
Total	50		100	
Class	Observed / Expected		Observed / Expected	
Parentals	31	25	62	50
Recombinants	19	25	38	50

TABLE 5.1 Critical Chi Square Values							
Degrees of Freedom	<i>p</i> Values						
	Cannot Reject the Null Hypothesis				Null Hypothesis Rejected		
	0.99	0.90	0.50	0.10	0.05	0.01	0.001
χ^2 calculations							
1	—	0.02	.45	2.71	3.84	6.64	10.83
2	0.02	0.21	1.39	4.61	5.99	9.21	13.82
3	0.11	0.58	2.37	6.25	7.81	11.35	16.27
4	0.30	1.06	3.36	7.78	9.49	13.28	18.47
5	0.55	1.61	4.35	9.24	11.07	15.09	20.52

χ^2 values that lie in the yellow-shaded region of this table allow you to reject the null hypothesis with > 95% confidence, and for recombination experiments, to postulate linkage.

答：

7. (From H3e, 5-10) In a particular human family, John and his mother both have brachydactyly (短指/趾畸形) (a rare autosomal dominant causing short fingers). John's father has Huntington Disease (亨廷顿舞蹈症) (another rare autosomal dominant). John's wife is phenotypically normal, and is pregnant. Two-thirds of people who inherit the *Huntington* (*HD*) allele show symptoms by age 50, and John is 50 and has no symptoms. Brachydactyly is 90% penetrant.
- What are the genotypes of John's parents?
 - What are the possible genotypes and their probabilities for John?
 - What is the probability the child will express both brachydactyly and Huntington Disease by age 50 if the two genes are unlinked?
 - If these two loci are 20 m.u. apart, how will it change your answer to part c?

答：